TW-1 Solar-Powered LoRaWAN® Field Sensor with Air, Soil and Rain Inputs

1 Features

- **Maintenance-free solar power**
  - 2 W panel on a ventilated sun roof
  - Protected 18650 Li-ion, 3 350 mAh
  - 251 days with no sun (1-min reporting)
- **Ultra-low power**
  - 7.2 µA sleep, whole system (typ)
  - 76 µA average at 5-min / SF9
- **LoRaWAN 1.0.3 Class A**, US915 sub-band 2
  - +20 dBm conducted, tuned π-match
  - 13.2 km line of sight at SF10 (tested)
- **Sensors**
  - Air temperature ±0.2 °C, humidity ±1 %RH
  - Capacitive soil moisture, 0.2 mm tipping-bucket rain
  - Internal T/RH for leak and condensation alerts
- **Built for sun and rain**
  - Cell ≤ 37 °C in Hawai'i sun (measured)
  - 1 m / 30 min immersion; UV-stable ASA
  - ±8 kV contact ESD on every field port
- Sealed, magnet-tap wake; open KiCad, firmware and CAD

2 Applications

- Irrigation scheduling and soil-moisture networks
- Frost and heat alerts for nurseries and orchards
- Rainfall networks and watershed monitoring
- Remote-site and shade-house environment logging

3 Description

The TW-1 is a complete, pole-mounted field sensor. It measures air temperature and humidity inside a louvred radiation shield, soil moisture and rainfall, and its own battery, solar and enclosure health. It reports over LoRaWAN every 5 minutes, or every minute in rapid mode, to any LoRaWAN 1.0.3 network server.

A linear charger with input-voltage regulation holds the 5 V panel near its maximum power point without an MPPT controller. A 75 nA buck-boost converter supplies the 3.3 V bus across the full Li-ion range. The panel mounts on a roof 25 mm above the white ASA enclosure and shades it, so the cell stays well inside its charging temperature range in tropical sun.

Table 3-1. Device Information

PART NUMBER	ENCLOSURE	BODY SIZE (NOM)
TW-1-US915	Pole-mount, IP67-tested	122 × 98 × 45 mm
TW-1-US915-PCBA	Board only	96 × 70 × 23 mm



TW-1 installed (sun roof, enclosure, radiation shield)

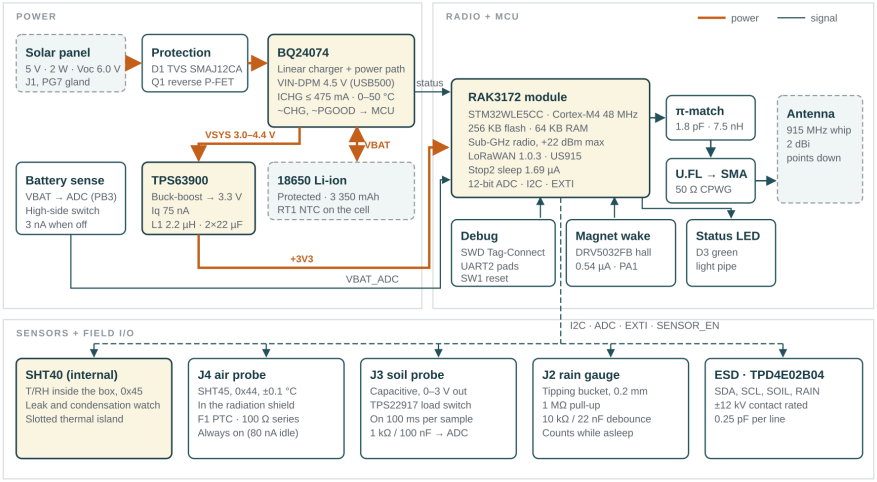


Figure 3-1. Simplified Block Diagram

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Revision History

Table R-1. Revision History

DATE	REVISION	NOTES
October 2026	*	Advance information. Hardware rev B, firmware v1.0.

4 Device Comparison

Table 4-1. Device Comparison

ORDERABLE	REGION / BAND	CONTENTS	MAX TX	STATUS
TW-1-US915-K	US915, sub-band 2	Sensor, roof + 2 W panel, radiation shield + SHT45 probe, soil probe, bracket	+20 dBm	Pilot
TW-1-US915	US915	Sensor, roof + panel, bracket (bring your own probes)	+20 dBm	Pilot
TW-1-US915-PCBA	US915	Assembled, tested board with cell holder	+20 dBm	Pilot
TW-1-AU915	AU915	As -K	+20 dBm	Planned
TW-1-EU868	EU868 (1 % duty cycle)	As -K	+14 dBm	Planned

ADVANCE INFORMATION. This document describes a product in pilot production. Characteristic data are representative values from the reference-design build and are subject to change without notice.

Supply and inputs at a glance

Table 4-2. Key Specifications

PARAMETER	VALUE	PARAMETER	VALUE
Sleep current (typ)	7.2 µA	Air temperature accuracy	±0.2 °C
Average, 5 min SF9	76 µA	RH accuracy (10–90 %)	±1 %RH
Dark autonomy (1 min)	251 d	Rain resolution	0.2 mm
TX power (conducted)	+19.9 dBm	Operating temperature	−10 to +60 °C
Range, SF10, LOS	≥ 13 km	Mass, box unit	292 g

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5 Connector Configuration and Functions



Figure 5-1. TW-1 PCBA, Top View (J1–J4 face the gland wall)

Table 5-1. Connector Functions

REF	PIN	NAME	TYPE[1]	DESCRIPTION
J1	1	SOLAR+	P	Panel positive, 0–10 V. Reverse-polarity protected (Q1), TVS clamped (D1).
J1	2	GND	G	Panel return
J2	1	RAIN	I	Rain-gauge reed to GND. 1 MΩ pull-up to 3.3 V, 10 kΩ / 22 nF debounce. Counted in sleep.
J2	2	GND	G	Ground
J3	1	3V3_SW	O	Switched 3.3 V for the soil probe, on 100 ms per sample. TPS22917 load switch.
J3	2	GND	G	Ground
J3	3	SOIL	AI	Probe output 0–3.0 V to 12-bit ADC (1 kΩ / 100 nF); 1 MΩ pull-down when unplugged
J4	1	3V3_EXT	O	Always-on 3.3 V for the air probe; 100 mA PTC (F1)
J4	2	GND	G	Ground
J4	3	SDA	I/O	I²C data (probe 0x44); 100 Ω series, ESD-protected
J4	4	SCL	I/O	I²C clock; 100 Ω series, ESD-protected
J5	—	RF	RF	U.FL, 50 Ω, to the SMA bulkhead and 915 MHz whip antenna
J6	1–6	SWD	I/O	Tag-Connect TC2030-NL: VCC, SWDIO, NRST, SWCLK, GND (pin 6 NC)
J7	1–4	UART	I/O	1.27 mm pads: GND, RX, TX, 3V3. AT commands and bootloader (BOOT0 high)
BT1	+/–	VBAT	P	Protected 18650 Li-ion only, 3.0–4.2 V
SW1	—	RESET	I	Reset button (inside the case)

(1) P = power, G = ground, I = input, O = output, AI = analog input, I/O = bidirectional, RF = radio frequency.
J1–J4: JST PH 2.0 mm side-entry. Mating cables pass through PG7 glands (cable Ø 3–6.5 mm).

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6 Specifications

6.1 Absolute Maximum Ratings

Table 6-1. Absolute Maximum Ratings

		MIN	MAX	UNIT
Solar input	J1 voltage (continuous)	−20	28	V
	J1 surge (8/20 µs)		±400	W
Battery	BT1 voltage	0	4.3	V
Field I/O	RAIN, SOIL, SDA, SCL	−0.3	3.6	V
Outputs	3V3_EXT + 3V3_SW load		100	mA
Temperature	Storage (Li-ion limited)	−20	60	°C

Stresses beyond those listed may cause permanent damage. These are stress ratings only; functional operation at these conditions is not implied.

6.2 ESD Ratings

Table 6-2. ESD Ratings

	STANDARD	VALUE	UNIT
$V_{(ESD)}$ on J1–J5	IEC 61000-4-2 contact discharge (criterion A)	±8 000	V
	IEC 61000-4-2 air discharge (criterion B on J4)(1)	±15 000	V

(1) One self-recovering reset at ±15 kV air on J4 (9 s to rejoin, no data loss). All other ports met criterion A.

6.3 Recommended Operating Conditions

Table 6-3. Recommended Operating Conditions

		MIN	NOM	MAX	UNIT
Panel	Open-circuit voltage at 0 °C	5.0	6.0	10	V
	Rated power	0.5	2	5	W
Cell	Protected 18650 Li-ion capacity	2 000	3 350	3 500	mAh
T_A	Ambient, operating	−10	25	55	°C
Interval	Uplink period (US915, SF7–SF10)	1	5	60	min

6.4 Environmental and Mechanical Characteristics

Table 6-4. Environmental and Mechanical

PARAMETER	TEST CONDITIONS	VALUE
Ingress	Immersion 1 m, 30 min / IPX5 hose	No ingress / pass (IP66 jet pending)
Thermal cycling	−10 ↔ +60 °C, 50 cycles	Pass, gasket set 11 %
UV resistance	500 h UV-A, white ASA	ΔE 1.7
Drop	26 × 1 m on concrete	Pass (cosmetic only)
Cell temperature in sun	Field, Sept, air max 31.9 °C	37.1 °C max
Enclosure	Box W × H × D / installed envelope	122 × 98 × 45 mm / 285 × 362 × 149 mm
Mass	Box unit / installed	292 g / 952 g

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6.5 Electrical Characteristics

T_A = 25 °C, VBAT = 3.7 V, 2 W panel, firmware v1.0, unless otherwise noted. Typical values are means of 10 rev B units.

Table 6-5. Electrical Characteristics

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
SYSTEM SUPPLY					
Sleep current, whole system	VBAT = 3.7 V, no sun, RTC running		7.2	10	µA
	VBAT = 4.2 V		7.6	8.7	µA
Average current	5-min interval, SF9, +20 dBm		76	100	µA
	1-min interval, SF9		351		µA
Charge per uplink	SF9, 11-byte payload, incl. RX windows		20.6		mC
+3V3 output voltage	I_{LOAD} = 10 mA	3.290	3.300	3.315	V
+3V3 ripple	I_{LOAD} = 10 mA, 20 MHz BW		18	21	mV_{PP}
+3V3 transient	5 → 87 mA step (TX start)		−40	±99	mV
Converter efficiency	VIN 3.7 V, I_{OUT} = 10 µA		88		%
	I_{OUT} = 10 mA		94		%
	I_{OUT} = 100 mA		93		%
SOLAR CHARGER					
Input regulation (VIN-DPM)	Panel current-limited	4.35	4.50	4.63	V
Input current limit	USB500 mode	450	474	500	mA
Charge regulation voltage		4.18	4.20	4.22	V
Termination current			10		% I_{CHG}
Charge window, hot	Cell temperature rising		50.4		°C
Charge window, cold	Cell temperature falling		-0.6		°C
Input overvoltage	J1 rising	10.2	10.5	10.8	V
Battery drain, no sun	BQ24074 BAT leakage		4.3	6	µA
Daily harvest	Clear / cloudy September day		2,336 / 862		mAh

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Table 6-6. Electrical Characteristics (continued)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
RADIO					
Uplink frequency	US915 sub-band 2, 125 kHz	902.3		914.9	MHz
Conducted TX power	At J5, 64 channels, 2 units	19.5	19.9	20.5	dBm
Return loss at J5 (installed)	915 MHz, lid closed, rev B match		-33	−15	dB
	Worst case 902–928 MHz		-19	−10	dB
Airtime, 24-byte frame	SF7 / SF9 / SF10, BW 125 kHz		62 / 206 / 371		ms
Range, line of sight	Gateway at 10 m, ≥ 50 % PDR, SF7		8.8		km
	SF10 (last test point)	13.2			km
OTAA join time	Median, SF9		6.0		s
SENSORS					
Air temperature accuracy	External SHT45 probe, −10 to +60 °C	−0.3	±0.1	0.3	°C
Air humidity accuracy	10–90 %RH, 25 °C	−2	±0.7	2	%RH
Internal temperature offset	Charging at 400 mA (self-heating)		0.94		°C
Soil input range	J3 pin 3	0		3.0	V
Soil ADC resolution			12		bit
Probe warm-up	3V3_SW on before sampling		100		ms
Rain pulse width	Reed closure	10			ms
Rain count rate, exact				5	tips/s
Battery voltage accuracy	ADC via 1:2 divider, 3.0–4.2 V		±15		mV
TIMING					
Awake time per uplink	Sense + TX + RX windows, SF9		2.19		s
Magnet wake	Tap (send now) / hold (rejoin)		0.2 / 3		s
Hall sensor sampling	DRV5032FB		5		Hz

Accuracy figures include the probe and the reading chain, measured against reference instruments (see the engineering report, section 10).

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6.6 Typical Characteristics

At T_A = 25 °C, VBAT = 3.7 V, unless otherwise noted.

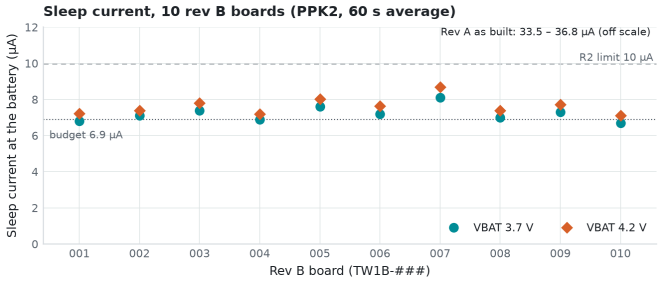


Figure 6-1. Sleep Current, 10 Units

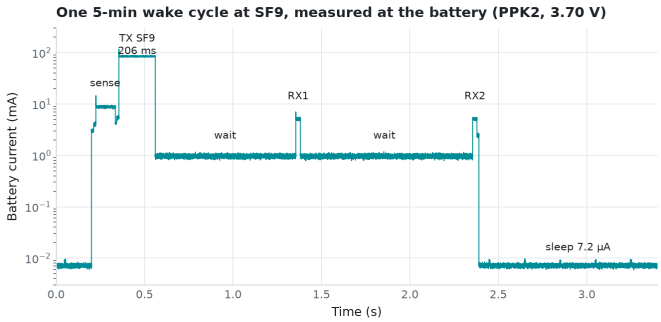


Figure 6-2. Supply Current Through One Uplink

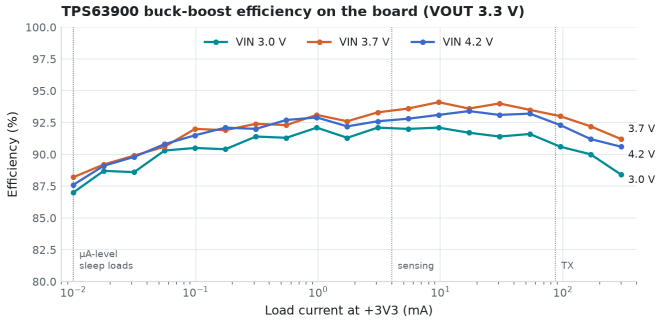


Figure 6-3. Converter Efficiency vs Load

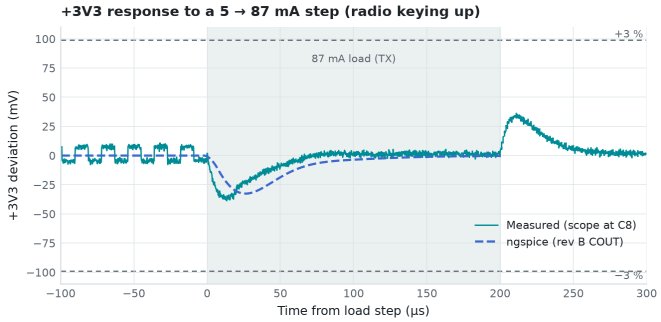


Figure 6-4. +3V3 Response to TX Load Step

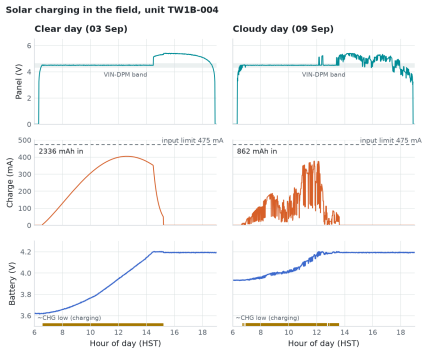


Figure 6-5. Solar Charging, Clear and Cloudy Days

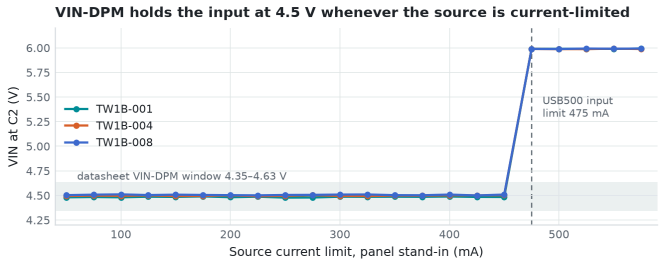


Figure 6-6. Input Regulation vs Panel Current

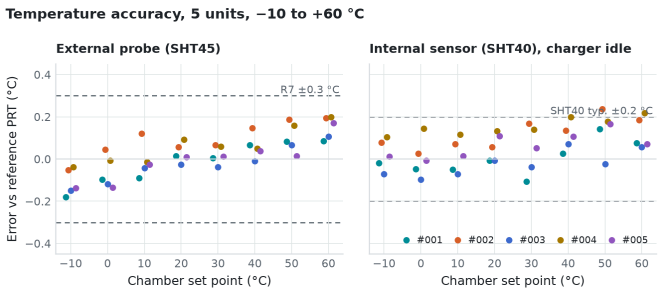


Figure 6-7. Air Temperature Error vs Temperature

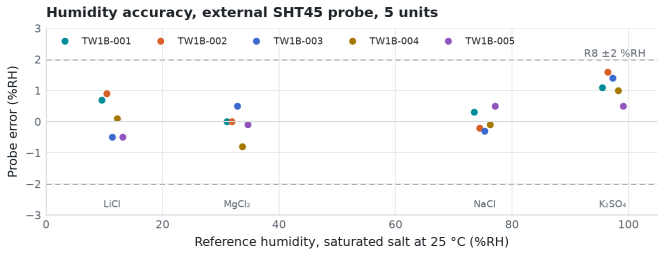


Figure 6-8. Humidity Error at Reference Points

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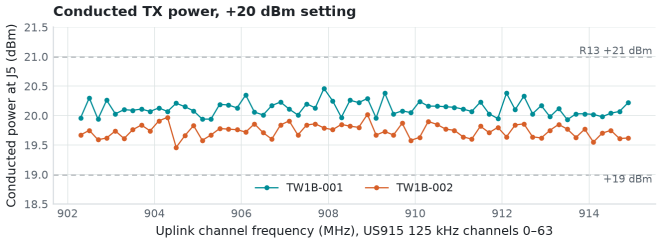


Figure 6-9. Conducted TX Power vs Channel

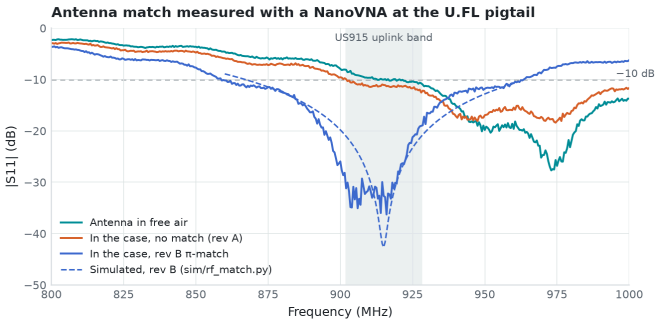


Figure 6-10. Return Loss at J5, Installed

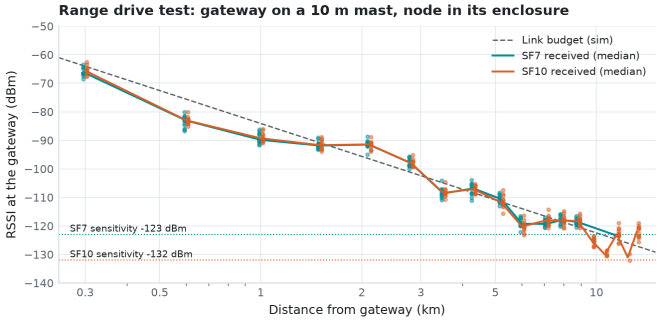


Figure 6-11. Packet Delivery vs Distance

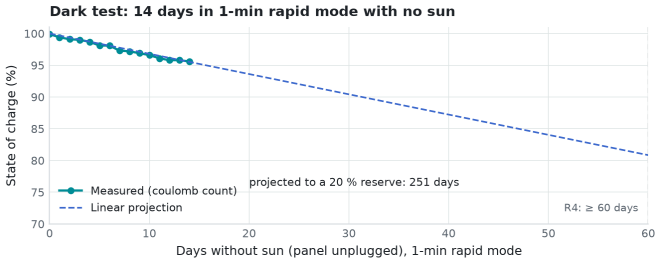


Figure 6-12. Battery State, No Sun, 1-Min Mode

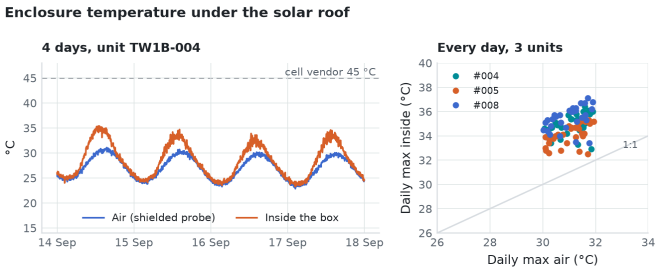


Figure 6-13. Enclosure vs Air Temperature, Field

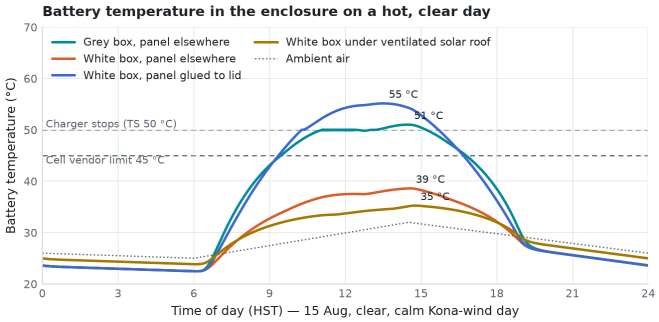


Figure 6-14. Cell Temperature vs Mounting (Model)

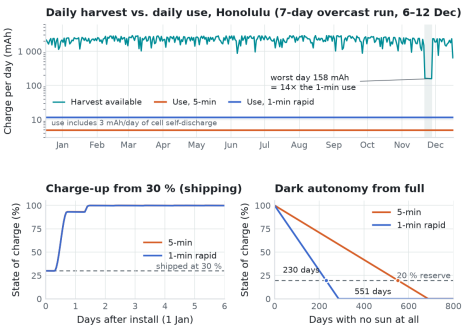


Figure 6-15. Battery State over a Year (Model)

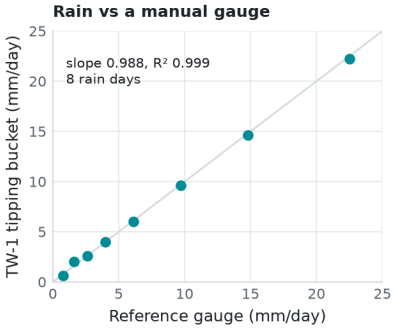


Figure 6-16. Rainfall vs Reference Gauge

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7 Detailed Description

7.1 Overview

The TW-1 combines a RAK3172 LoRaWAN module (STM32WLE5CC with a sub-GHz radio), a BQ24074 linear charger with power path, a TPS63900 buck-boost converter, a protected 18650 Li-ion cell and four field inputs on one 4-layer board inside a sealed ASA enclosure. Firmware runs on the module (RAKwireless RUI3). The device sleeps in STOP2 between scheduled uplinks and wakes early on rain-gauge pulses and magnet taps.

7.2 Functional Block Diagram

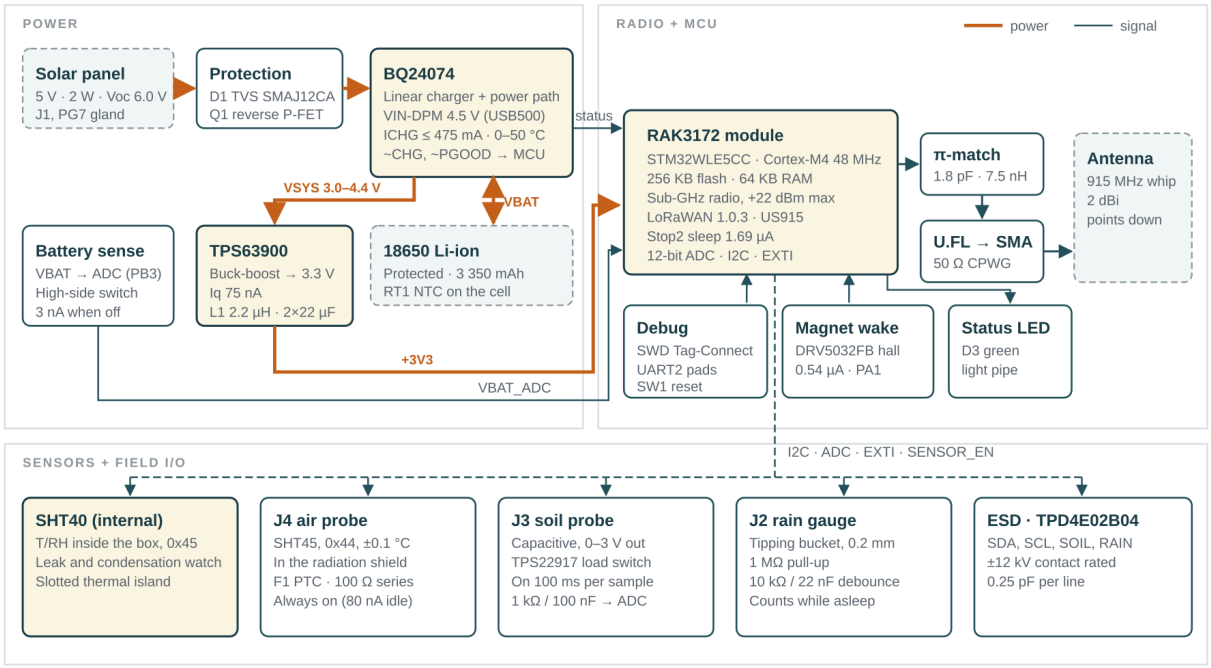


Figure 7-1. Functional Block Diagram

7.3 Feature Description

7.3.1 Solar Power Path

The charger runs in USB500 mode, so its input-voltage loop (VIN-DPM, 4.50 V typ) cuts the charge current before the panel collapses. A 5 V panel's maximum-power voltage falls to about 4.36 V at a working temperature of 55 °C, so this fixed point captures 99 % of the available power without an MPPT controller. The system load always runs from VSYS; the cell supplies only the shortfall.

7.3.2 Temperature-Qualified Charging

A 10 kΩ NTC against the cell stops charging outside about 0 to 50 °C. The ventilated roof keeps the cell well below the hot limit: 35 °C modelled on a calm August day, 37 °C measured in the field.

7.3.3 Low-Power Sensing

The soil probe is powered only while sampling. The battery divider uses a high-side switch that leaks 3 nA when off. The rain input is filtered in hardware (τ 0.22 ms on close, 22 ms on release) and counted by interrupt while the MCU sleeps.

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7.4 Device Functional Modes

Table 7-1. Functional Modes

MODE	ENTERED BY	SUPPLY CURRENT	DURATION
Sleep	End of an uplink cycle	7.2 µA	Until the next deadline or an interrupt
Measure	RTC alarm	≈ 9 mA	150 ms (incl. 100 ms probe warm-up)
Transmit	After measure	87 mA	62–371 ms (SF7–SF10)
Receive	RX1 / RX2 windows	5.2 mA	2 × ≈ 25 ms
Rapid	Downlink 0x02, rain rate or soil change	351 µA avg	1-min interval, timed
Join	First boot, or magnet held 3 s	—	6 s median, exponential back-off

7.5 Programming

7.5.1 Uplink Format

Table 7-2. fPort 1 Measurement Frame (11 bytes, big-endian)

BYTE	FIELD	TYPE	SCALE	RANGE
0	Flags	u8	b0 rapid, b1 charging, b2 power good, b3 battery low, b4 air OK, b5 soil OK, b6 internal OK	
1–2	Air temperature	s16	0.01 °C	0x7FFF = missing
3	Air humidity	u8	0.5 %RH	0–100 %
4	Soil moisture	u8	0.5 %	0–100 %
5–6	Rain tips	u16	0.2 mm / tip	Cumulative, wraps
7	Battery	u8	2 500 mV + 10 mV·n	2.50–5.05 V
8	Internal temperature	s8	0.5 °C	0x7F = missing
9	Internal humidity	u8	0.5 %RH	
10	Panel voltage	u8	25 mV	0–6.37 V

7.5.2 Downlink Commands (fPort 10)

Table 7-3. Downlink Commands

BYTES	COMMAND	NOTES
`01 HH LL`	Set interval (s)	60–3 600 s
`02 MM`	Rapid mode for MM minutes	02 00 ends rapid mode
`03`	Send health frame now	fPort 2, confirmed
`04 B0 07`	Reboot	Two-byte key guards against stray bytes

Decoders for The Things Stack and ChirpStack v4: `firmware/decoder/ttn_decoder.js`.

8 Application and Implementation

Note: Information in the following application sections is not part of the specification. Customers validate and test their own installation.

8.1 Typical Application: Irrigation and Frost Monitoring



Figure 8-1. Typical Installation

Table 8-1. Design Requirements

PARAMETER	EXAMPLE VALUE
Reporting	5 min; rapid 1 min during irrigation
Gateway distance	0.5–5 km, one gateway per farm
Probe height	1.5–2 m air; soil probe at root depth
Panel orientation	Facing south, 20° tilt (Hawai'i)
Network	ChirpStack or The Things Stack

8.2 Detailed Installation Procedure

1. Mount the bracket on a 25–50 mm pole, box 1.5–2 m above ground. Keep it 3 m or more from metal roofs and water tanks to limit RF shadow.
2. Fit the probe cables through the PG7 glands with a drip loop below each gland, then torque the gland domes to 2.5 N·m.
3. Insert a protected 18650 (ships at about 30 % charge), close the lid and torque the M4 screws to 1.2 N·m in a cross pattern.
4. Tap the magnet on the lid logo. The LED blinks while joining and holds solid for 2 s when joined.
5. Check the first uplinks in the network server: RSSI above −115 dBm at SF9 gives about 10 dB of fade margin.

8.3 Application Curves

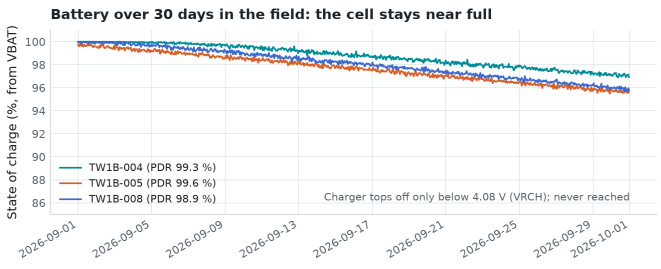


Figure 8-2. Battery State, 30-Day Pilot (3 Units)

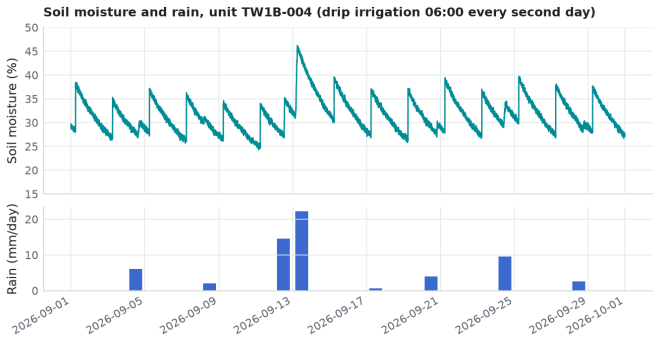


Figure 8-3. Soil Moisture and Rain, Pilot Site

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9 Power Supply Recommendations

- **Panel:** 5 V class (10 series cells), V_{OC} ≤ 6.5 V at 0 °C, 0.5–5 W. A 6 V class panel works, but wastes 10–20 % of its power, because input regulation holds it at 4.5 V.
- **Cell:** protected 18650 Li-ion only (PCM cut-off ≥ 2.5 V). Unprotected cells can be deep-discharged after very long dark periods.
- **Energy margin:** on the worst simulated day in Honolulu the panel still delivers 158 mAh, 14× the 1-minute-mode use. With no sun at all, a full cell lasts 251 days in rapid mode and about 551 days at 5 minutes.
- **Storage:** ship and store at 30–50 % charge, 10–25 °C. The device stays joined to the network while stored, so expect one uplink per interval.

10 Installation

10.1 Guidelines

- Antenna vertical, pointing down (as shipped), clear of the pole by 40 mm or more. Do not paint the antenna or the lid.
- Radiation shield in free air: not over pavement or above a roof, and 1 m or more from walls.
- Keep the panel unshaded from 9 am to 3 pm. In shade the device still runs for months on the cell, but charges slowly.
- Coastal sites (within about 200 m of surf): use the stainless hardware kit and inspect the glands yearly. Salt-fog qualification is pending.

10.2 Layout Example (PCBA)

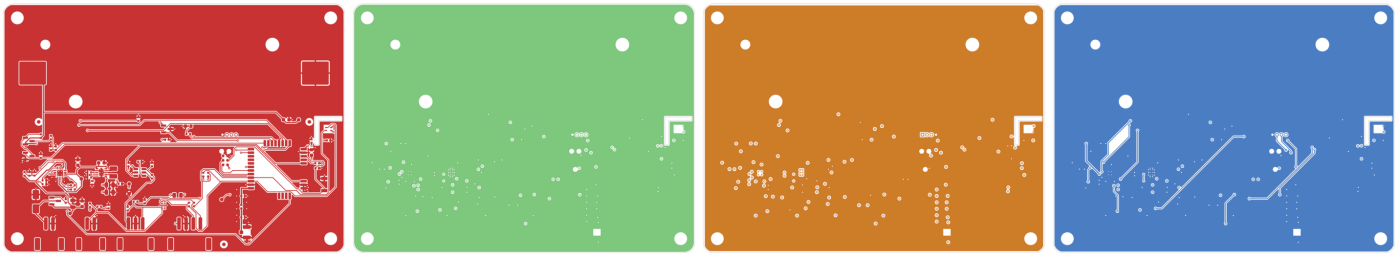


Figure 10-1. Copper Layers L1–L4: Signals, GND Plane, 3.3 V Plane, Signals

The PCB is 4-layer, 1.6 mm (JLC 7628 stackup), with 50 Ω CPWG on L1 (0.35 mm trace, 0.20 mm gap) and a fenced RF path. The SHT40 sits on a slotted thermal island. KiCad 9 source, Gerbers and BOM: github.com/binaryBeachLLC/tradewind.

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11 Mechanical, Packaging, and Orderable Information

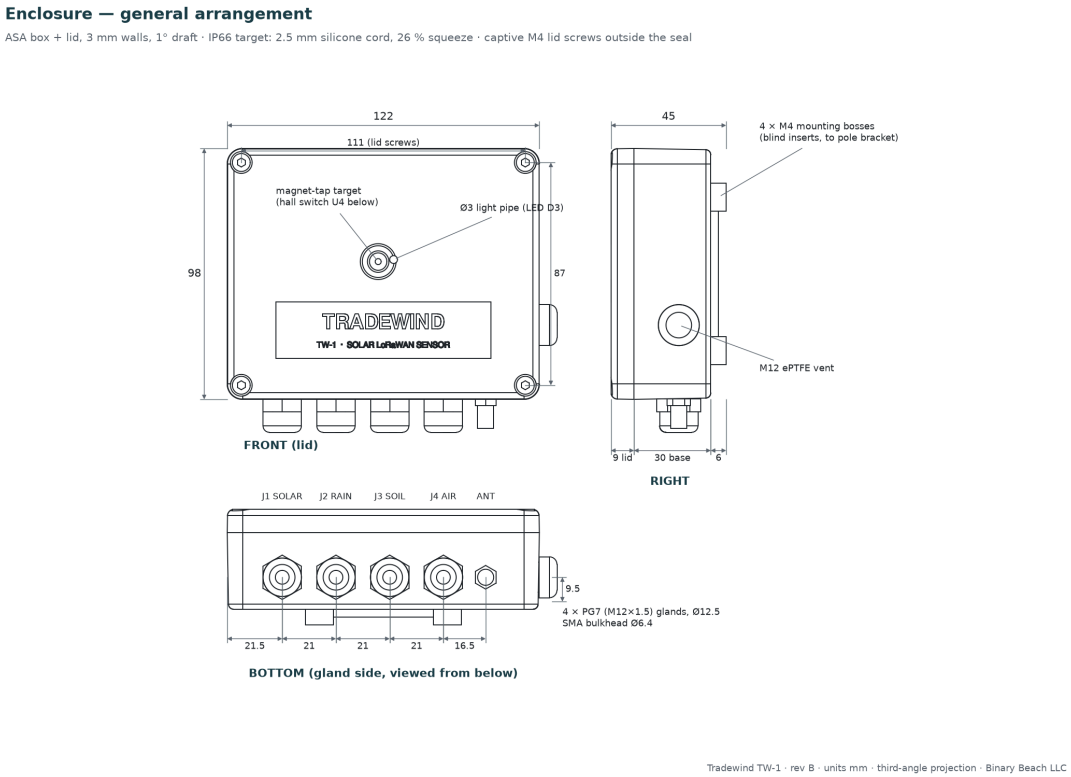


Figure 11-1. Enclosure Outline Drawing (dimensions in mm)

Table 11-1. Orderable Information

ORDERABLE PART	STATUS	PACKAGE	PACK QTY	CONTENTS
TW-1-US915-K	PILOT	Kit box	1	Sensor, roof + panel, shield + probe, soil probe, bracket, 2 × U-bolt, magnet key
TW-1-US915	PILOT	Kit box	1	Sensor, roof + panel, bracket, magnet key
TW-1-US915-PCBA	PILOT	ESD bag	10	Tested boards (no cell)

Table 11-2. Mechanical Data

ITEM	MATERIAL / FINISH	VALUE
Box and lid	ASA, white, UV-stabilized	3 mm wall, 1° draft
Seal	Silicone cord, 2.5 mm	26 % compression
Hardware	A2 stainless; brass inserts	M4 lid, M3 PCB
Cable entries	PG7 glands, PA66	4 × (Ø 3–6.5 mm)
Antenna	915 MHz whip, SMA	2 dBi, about 195 mm

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12 Device and Documentation Support

12.1 Documentation

- *Tradewind TW-1 Engineering Design & Test Report*, BB-ENG-2026-014: design analysis, simulation, layout, bring-up and characterization.
- Design files: KiCad 9 project, Gerbers, BOM, enclosure CAD (STEP/STL), firmware, decoders and test logs: github.com/binaryBeachLLC/tradewind.
- Datasheets of the main components: RAK3172 (RAKwireless), BQ24074, TPS63900, TPS22917, TPD4E02B04, DRV5032 (Texas Instruments), SHT4x (Sensirion).

12.2 Regulatory

The radio is a modular-certified RAK3172 (contains FCC ID and IC as marked on the module). The host product requires FCC Part 15 Subpart B verification and labeling before sale; a pre-scan is planned with the pilot build.

12.3 Trademarks

LoRa® and LoRaWAN® are trademarks of Semtech Corporation or its subsidiaries. All other trademarks are the property of their respective owners.

12.4 Electrostatic Discharge Caution

> The bare PCBA can be damaged by ESD. Handle it at a grounded workstation; the field ports are protected only once the board is installed in the enclosure.

12.5 Glossary

TERM	MEANING
VIN-DPM	Input-voltage dynamic power management: the charger reduces its input current to hold the panel voltage
SF	LoRa spreading factor (higher means longer range, longer airtime)
PDR	Packet delivery ratio
CPWG	Grounded coplanar waveguide
OTAA	Over-the-air activation (LoRaWAN join)

12.6 Important Notice

Binary Beach LLC provides this datasheet and the reference design "as is", for evaluation and engineering use. Advance-information data are representative of the rev B pilot build and are subject to change. Binary Beach makes no warranty of fitness for a particular purpose; users are responsible for validating the design in their application and for meeting the regulations that apply where it is used. Hardware design files are licensed under CERN-OHL-P-2.0 and firmware under the MIT License.

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